A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 33

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients – worms create tunnels improving drainage and aeration **up to 2** Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds **up to 2** Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions – nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption **up to 2** Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ammonification** → ammonium released from dead organic matter and protein • **humus** → dark, decomposed organic matter in soil • **peds** → soil crumbs / aggregates of soil particles	6

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** This is the range of tolerance of most plants and soil biota – acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes **1** In alkaline soils phosphates become insoluble / unavailable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate **1** Infiltration falls so waterlogging is more likely and runoff / erosion increase **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Interstitial** → pore spaces / air spaces between the particles • **nitrification** → ammonium oxidised to nitrite and then nitrate • **Infiltration** → water soaking into the soil surface	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated **1 Grit traps:** the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	2
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge **1** This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter **1** Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process **1** The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge' – returning them maintains a large bacterial population so digestion is rapid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria **1** Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) **1** Phosphate stripping with iron(III) sulfate – micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1 Topography:** a narrow exit from a large deep basin so a small dam holds a large volume **1 Geology:** impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity **1 Catchment area:** large enough with regular, reliable rainfall / river inflow – climate not too hot (evaporation losses) **1 Catchment land use:** low pollution risk (industry, pesticides) – low soil erosion (sedimentation reduces capacity) **1 Infrastructure / proximity to demand:** access for construction, cost of transporting water **1 Existing land use:** value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
03.2	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2 Advantage:** fewer land-use conflicts as no land is flooded – large volume near coastal cities **up to 2 Disadvantages:** destroys / changes important intertidal habitats (mudflats for wading birds) – obstacle to shipping – collects pollution from the whole river catchment – sedimentation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Very energy-intensive and expensive (reverse osmosis / distillation) **1** Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	1 mark for the point below. **1** Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **infiltration** → water soaking into the soil surface **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	
04.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Ground surface:** pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals – trapped predators eat others – preservative attracts / repels – escapes **2 Leaf litter:** Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light – some die or stop moving – large ones cannot pass the mesh **2 Branches:** beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour – some species not dislodged / fly away – high branches unreachable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pitfall trap** → a sunken container that catches ground invertebrates • **Tullgren** → a funnel that drives soil invertebrates down into a collecting jar	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught **1** Brightness / wavelength of the lamp affects attractiveness so catches vary between traps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** The area of ground covered by the suction pipe is known so invertebrates per m^2 can be calculated **1** A pooter is only a handling / collection device for individuals already found, not a sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** The human ear is sensitive to a very wide range of sound pressures / powers (about 10^{12} range) **1** A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Difference = 30 dB = 3 x 10 dB **1** = 10^3 = 1000 times **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** 88 dB **1** Doubling the sound intensity increases the level by 3 dB (logarithmic scale) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	1 mark for the point below. **1** 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing – quieter sounds exist but cannot be heard **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **threshold** → the concentration below which no harm is detected	1
05.5	**1 mark each** 1 mark for each of the 3 points below. **1** 20 to 20 000 Hz **1** dB(A) weights sounds between about 1000 and 4000 Hz more heavily **1** Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance – range narrows with age or noise damage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1 LEO:** high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover **1** But each area is only revisited every 15 days so they cannot follow rapidly changing weather **1 Geostationary:** constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
06.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1** Wind velocity **1** Ocean currents / sea surface temperature **1** Wave height **1** Ice thickness **1** Vegetation cover **1** Sea level **1** Atmospheric temperature / composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome **1** If the prediction matches, the model can be trusted (with caution) – it is continually modified as more data are collected and differences analysed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Models need data from all regions / oceans / altitudes – gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions **1** Unrepresentative data (eg urban heat islands, short time series) would bias the model – remote sensing helps fill gaps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **Critical Group** → the people most at risk / most heavily exposed	1
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate **1** No need to sample sites that were never likely to be polluted – sampling also checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify **1** So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
08.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 180 / 1.2 = 150 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
08.2	1 mark for the point below. **1** Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 4 points below. **1** Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium (1.2 vs 0.05 mg kg^{-1}) **1** Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years **1** Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) **1** Rice is the staple food so residents receive a large dose – village B has no exposure and no disease (correlation supports causation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together • **causation** → one variable actually causing the change in the other **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
08.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble **1** Cover / cap spoil heaps to prevent leaching and dust **1** Recycle nickel-cadmium batteries / hazardous waste landfill **1** Phytoremediation of contaminated paddy soils **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
08.5	**1 mark each** 1 mark for each of the 2 points below. **1** Manufacture may release cadmium in effluents / dust **1** Panels have a limited life – disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium **1** Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet **1** International agreement on ownership of the seabed and its resources is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Collectors disturb the seabed and kill benthic organisms living on it **1** Separating nodules from sediment creates plumes that increase turbidity with unknown consequences **1** Re-deposition of sediment covers and kills filter feeders and other benthic organisms – slow recovery in the deep sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.3	**1 mark each** 1 mark for each of the 4 points below. **1** Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) **1** Exploitable land deposits are limited and mainly in China – chemical separation is complex and produces large amounts of toxic waste **1 Bacterial adsorption:** dissolved rare earth ions adsorb onto the cell surfaces of bacteria (Bacillus subtilis) and are washed off at specific pHs to separate them **1** Deep sea sediments contain relatively high concentrations but no viable extraction method yet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Features:** renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts • **Fossil fuels:** high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO2), politically concentrated (Middle East oil) • **Nuclear:** very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition • **Renewables:** low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs • **Sustainability concepts:** impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion) • **Economic and political factors:** subsidies, guaranteed prices, planning, external costs not included in price, energy security • **Judgement:** a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable – transition must start before depletion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **sustainability** → meeting present needs without stopping future generations meeting theirs • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **base load** → the steady minimum demand that must always be met **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Whether a resource suits a use depends on its features: renewable or not, abundance, energy density, locational constraints, intermittency and predictability, the conversions needed, ease of storage and transport, how developed the technology is and its impacts. Fossil fuels dominate because they combine high energy density, easy storage and transport, high temperatures for industry and mature infrastructure - which is why petrol and diesel still monopolise transport. But they are finite, cause acid rain, smoke and CO_2, and are politically concentrated. Nuclear fuel has the highest energy density and gives reliable base load with few deliveries, yet the technology is complex and expensive and produces long-lived waste against public opposition.	
	Renewables have the opposite profile: low energy density needing large collectors, intermittency (wind and solar unpredictable, tidal predictable), strong locational constraints and no direct storage, so they need conversion, storage and grid links and cannot yet reach fossil-fuel temperatures or replace liquid fuels. Sustainability must count impacts before use - extraction, processing, embodied energy of equipment - as well as pollution, habitat damage and depletion during use, and economics and politics (subsidies, planning, external costs left out of prices, energy security) shape what is actually chosen.	
	No single resource is suitable for everything. Electricity, heat and transport have different requirements, so the most sustainable system is a diverse mix matched to each use with storage, and the transition must begin before depletion forces it.	
	Weaker answer - Level 1 (2 marks)	
	Energy resources have different features. Fossil fuels are non-renewable and will run out but they have a high energy density. Renewable energy like wind and solar will not run out but they are intermittent because the wind does not always blow. Nuclear power produces radioactive waste. Renewable energy is more sustainable because it does not produce carbon dioxide. So we should use more renewable energy.	
	What makes the difference	
	• Level 1 mentions two or three features; Level 3 works through the full set and links each to a specific use (transport, base load, industry).	
	• Level 3 includes sustainability concepts (impacts before and during use, external costs) and political factors that Level 1 omits entirely.	
	• The conclusion in Level 3 is a reasoned position (a diverse mix matched to uses); Level 1's 'use more renewables' does not follow from a discussion.	

Do not write outside the box

Question	Marking guidance	Marks
10.2	1 mark for the point below. **1** Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Stages of investigation:** identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • **Why Antarctica:** low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • **Effects of UVB:** skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • **Solutions:** Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • **Evaluation of success:** ozone hole area / depth stabilising, CFC levels falling	25
	– limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier – role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

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